

Agentic Integration Engine (AIE) for iFAMS modernization

AN AI-ENABLED APPROACH TO ACCELERATE INTERFACE DELIVERY

Executive Summary

The Department of Veterans Affairs manages one of the most expansive financial and technology ecosystems in government. Delivering seamless integration across this environment, particularly for initiatives like the Financial Management Business Transformation (FMBT) program, presents significant challenges. These include reducing interface development timelines, ensuring compliance with enterprise standards, executing accurate and timely data conversion from legacy systems, and minimizing duplication across systems.

As interface complexity grows, consistent governance and comprehensive validation remain essential to program success. Opportunities exist to expand risk-based testing coverage, enhance coordination of test environments, and better align delivery schedules with validation outcomes. In parallel, strengthening mapping completeness and increasing dry-run rigor in conversion activities can further reduce data integrity risk and improve reliability across integration waves.

While VA has adopted advanced tools to improve efficiency, current approaches lack the ability to automate end-to-end interface design and enforce data conversion compliance at scale. These limitations have contributed to audit findings around traceability gaps, weak

dependency management, and insufficient validation of financial data flows.

At Guidehouse, we see an opportunity to significantly accelerate delivery of FMBT interfaces and conversion processes while reducing delivery and quality risks. The Agentic Integration Engine (AIE) introduces secure, intelligent automation powered by advanced AI capabilities to accelerate interface delivery and streamline data migration for FMBT while remediating prior audit concerns. By connecting existing systems and applying reusable patterns, AIE enables teams to:

- Generate compliant interface specifications automatically, ensuring alignment with VA standards.
- Enforce traceability and dependency checks, reducing risk of undocumented changes.
- Validate financial data flows and conversion outputs against governance rules, strengthening audit readiness.

This approach reduces development time, improves quality, and closes documentation and process inconsistencies by embedding governance and validation directly into the integration lifecycle.

Solution Overview

The Agentic Integration Engine (AIE) is an intelligent orchestration framework designed to accelerate interface delivery while embedding governance, security, and auditability into every step. AIE transforms interface development from a manual, fragmented process into a governed automation layer that operates entirely within VA-approved boundaries and integration methods.

At its core, AIE combines two powerful components:

Knowledge Fabric: A structured, versioned repository of FMBT artifacts such as ICDs, RTMs, and test packs, enabling rapid retrieval, traceability mapping, and contextual recommendations.

Agentic AI Automation Layer: Specialized agents for requirements mapping validation, test planning, and reconciliation work in concert governed by the Model Context Protocol (MCP). MCP enforces secure context exchange, lineage tracking, and policy compliance across all workflows. This enables FMBT to accelerate interface delivery, improve quality through automated validation, and strengthen audit readiness by embedding compliance and traceability into every step.

AIE supports FMBT's environment by interacting only through VA-approved pathways:

- FUSE Enterprise Service Bus (ESB) and Momentum Web Services/APIs for near real-time exchanges, ensuring schema compliance and full validation evidence.
- Bulk Operations via ESB orchestrations and Momentum batch utilities for validated processing, reconciliation, and audit-ready packaging.

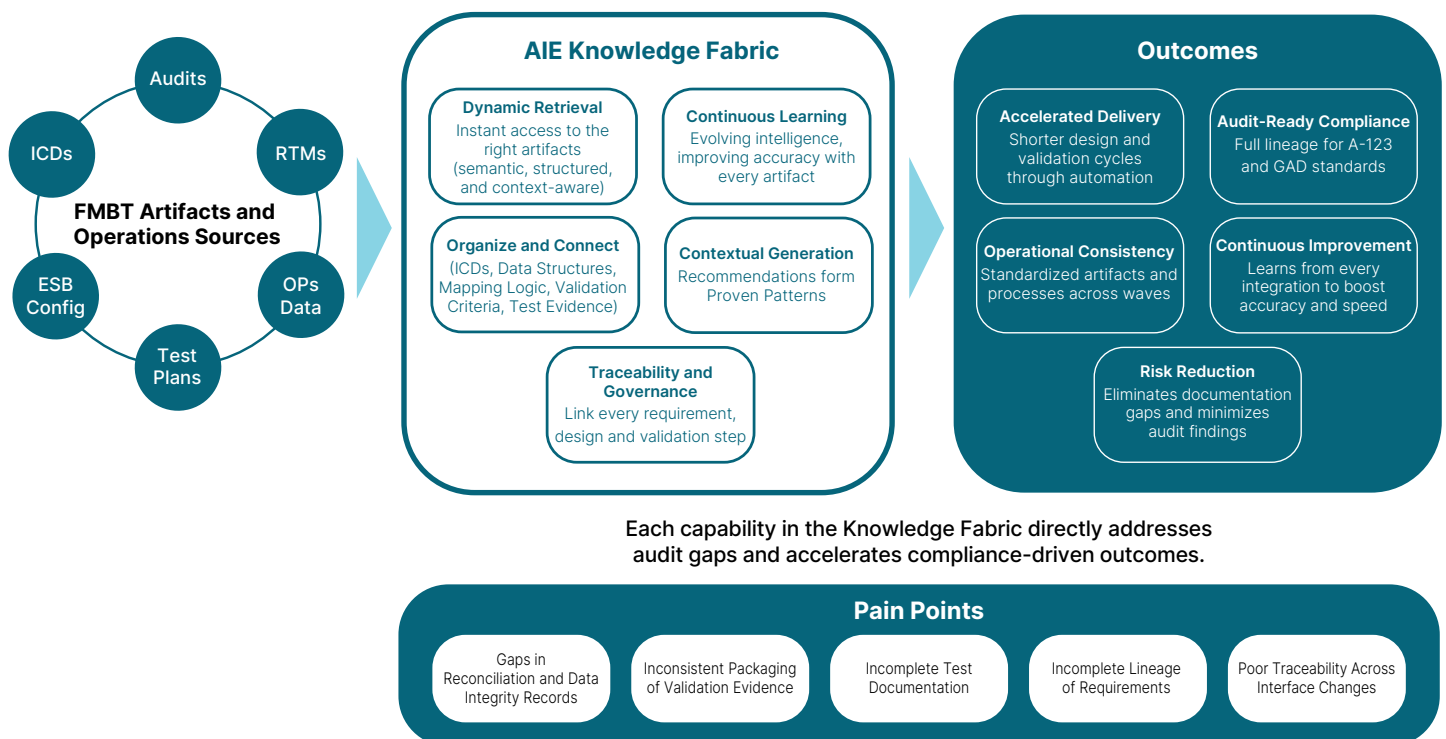
The AIE Knowledge Fabric—An evolving knowledge repository

The AIE Knowledge Fabric goes far beyond a traditional document archive; it is an intelligent, agent-driven framework that continuously adapts and improves by learning from past integration efforts and ongoing operations. Rather than merely storing information, it organizes and connects key integration components, such as interface definitions, data structures, mapping logic, validation criteria, and testing evidence, in a way that makes them easily discoverable and directly useful for future interfaces that share characteristics.

This fabric enables:

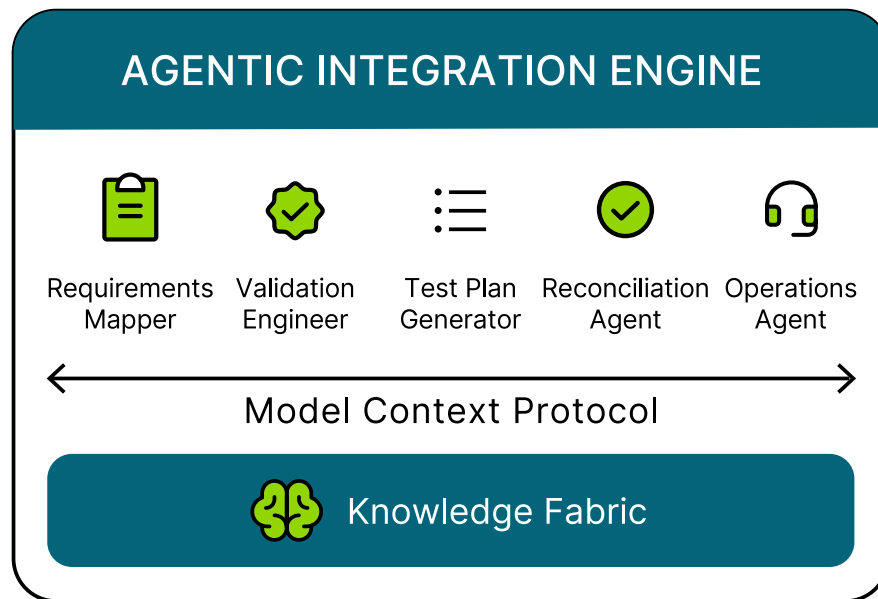
- **Dynamic Retrieval:** AI agents can query historical ICDs, RTMs, ESB route specs, test packs, and reconciliation records to inform design decisions in real time.
- **Contextual Generation:** When building new interfaces, the system generates recommendations and templates based on proven patterns from past waves.
- **Traceability & Governance:** Each entity is linked across requirements, design, and validation, ensuring compliance and reducing risk.
- **Continuous Learning:** As new artifacts are added, the fabric evolves, improving accuracy and accelerating integration with every wave.

Figure 1: AIE Knowledge Fabric Capabilities



Agentic Integration Engine—Accelerating interface delivery

Figure 2: Agentic Integration Engine



The Agentic Integration Engine (AIE) applies AI-powered agents orchestrated through the Model Context Protocol (MCP) to automate interface analysis, mapping, testing, reconciliation, and validation. These agents run as secure containerized services on VAPO, with inference provided entirely inside VAEC using FedRAMP High OpenAI services in Microsoft Azure Government and VAEC-hosted Retrieval-Augmented Generation (RAG) components (VA storage, vector search, and metadata indexing). All artifacts, embeddings, and evidence remain within VA boundary controls. The AIE generates validated interface configuration packages that feed into the existing FUSE interface release and approval process, accelerating delivery and improving audit readiness without modifying FUSE architecture or relying on any external AI endpoints.

By applying AI-driven orchestration, AIE automates critical steps often missed in traditional development. It transforms interface

delivery from manual and fragmented to governed and highly automated. AIE pre-populates ICD sections, recommends validated mappings, generates test scripts, and packages audit-ready evidence so developers receive complete, ready-to-implement specifications and compliance artifacts upfront.

Agentic RAG leverages the structured Knowledge Fabric to automate and accelerate interface delivery. By combining vector search and keyword indexing, agentic RAG quickly locates relevant artifacts, ICDs, RTMs, mappings, and test packs from prior integration waves. It generates draft interface documents with embedded lineage, enabling rapid human review while maintaining strong audit trails. All retrieval and generation run under MCP governance for secure context, traceability, and compliance. This streamlined approach delivers accurate, evidence-backed specifications at speed within VA-approved boundaries.

Conclusion

In summary, the Agentic Integration Engine delivers a transformative leap in interface development and data migration for the FMBT program. By securely automating critical integration tasks and maintaining rigorous governance, this solution dramatically reduces manual effort and risk while accelerating time to production. This end-to-end approach ensures that teams consistently receive complete, audit-ready specifications, empowering the FMBT team to deliver high-quality interfaces and auditable data migrations with greater speed, accuracy, and confidence.

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